

Homework 4

Release Date: 1404/8/25

Deadline: 1404/9/3

(*) starred problems are optional and have a bonus mark!

1 Pre-envelope and LPE for an SSB signals (18pts)

Let $m(t) = \text{sinc}^2(t)$ and let $\hat{m}(t)$ denote the Hilbert transform of $m(t)$. Consider the bandpass signal

$$x(t) = m(t) \cos(2\pi f_0 t) - \hat{m}(t) \sin(2\pi f_0 t).$$

- 1) Find the pre-envelope $z(t)$ of $x(t)$ and the corresponding low-pass equivalent (complex envelope) $g(t)$.
- 2) Determine and sketch the Fourier transform $X(f)$ of $x(t)$. What is the (null-to-null) bandwidth of $x(t)$?
- 3) Repeat parts 1 and 2 for

$$x(t) = m(t) \cos(2\pi f_0 t) + \hat{m}(t) \sin(2\pi f_0 t).$$

2 Weaver's SSB Modulator (20+5pts)

Consider the Weaver SSB modulator shown in Fig. 1. Assume the input is $x(t) = \cos(2\pi f_m t)$ with $f_m < W$. Show that, with appropriate choices of f_1 and f_2 , the output $y(t)$ is an SSB signal.

1. Determine conditions on f_1, f_2 , and the LPF bandwidth W so that the unwanted sideband is rejected.
2. State which sideband (USB or LSB) appears at the output for your choice of (f_1, f_2) . How would you obtain the other sideband?
3. For a general baseband $x(t)$ that is bandlimited to W Hz, prove that the same parameter choices produce an SSB signal centered at f_2 .
4. Sketch the spectra at the inputs and outputs of the mixers, at both LPF outputs, and at the final summer. Label all key frequencies.
5. (*) Derive $y(t)$ explicitly using trigonometric identities and show that one sideband cancels due to the 90° phase shifts.

5 VSB Modulation Design (10pts)

For a VSB amplitude modulation, with a modulating signal of bandwidth W rad/s:

1. Explain how the modulated signal is generated, and draw a block diagram with the different elements that make up the transmitter.
2. Indicate what condition the VSB filter has to meet for this type of modulation. For each of the filters in Fig. 2, state whether or not this condition is met and, if so, whether it would be used for upper- or lower-sideband modulation.

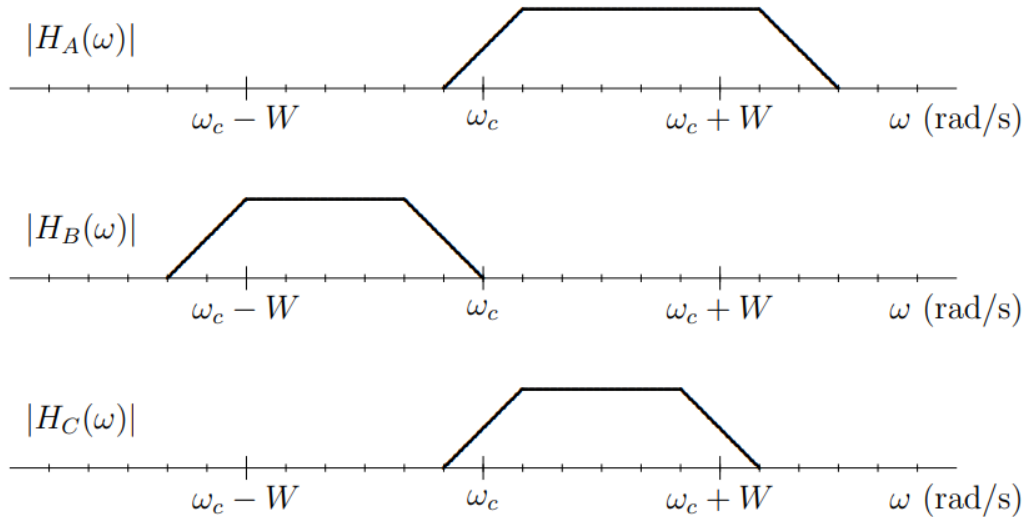


Figure 2: Filters to be evaluated for the VSB condition.

6 Interpreting Modulated Signals (20pts)

Four modulated signals are illustrated in Fig. 3.

1. Figures A, B, C, and D show the modulated signal when the same modulating signal $m(t)$ is transmitted using one of these amplitude modulation schemes: conventional AM (with modulation index $a = \frac{1}{2}$), double-sideband suppressed-carrier (DSB-SC) or something else. For each figure, identify which scheme it corresponds to and state its distinctive features (answers without justification will not be graded). If a figure cannot be identified from the given plot, explicitly say so and explain why Remark. To facilitate identification, the carrier $c(t)$ has been included in the four figures as a thin red line.
2. For each of the scenarios below, state which analog modulation you would use and justify your choice by the specific property that makes it appropriate, and what advantage it provides over other options (do not limit yourself to the 4 indicated figures in the previous section).
 - i) A system in which the main priority is that the receivers are as economical as possible.
 - ii) A single-cable multiplexing system with a 400 MHz bandwidth, in which it is desirable to use the capacity to carry the greatest possible number of voice signals.

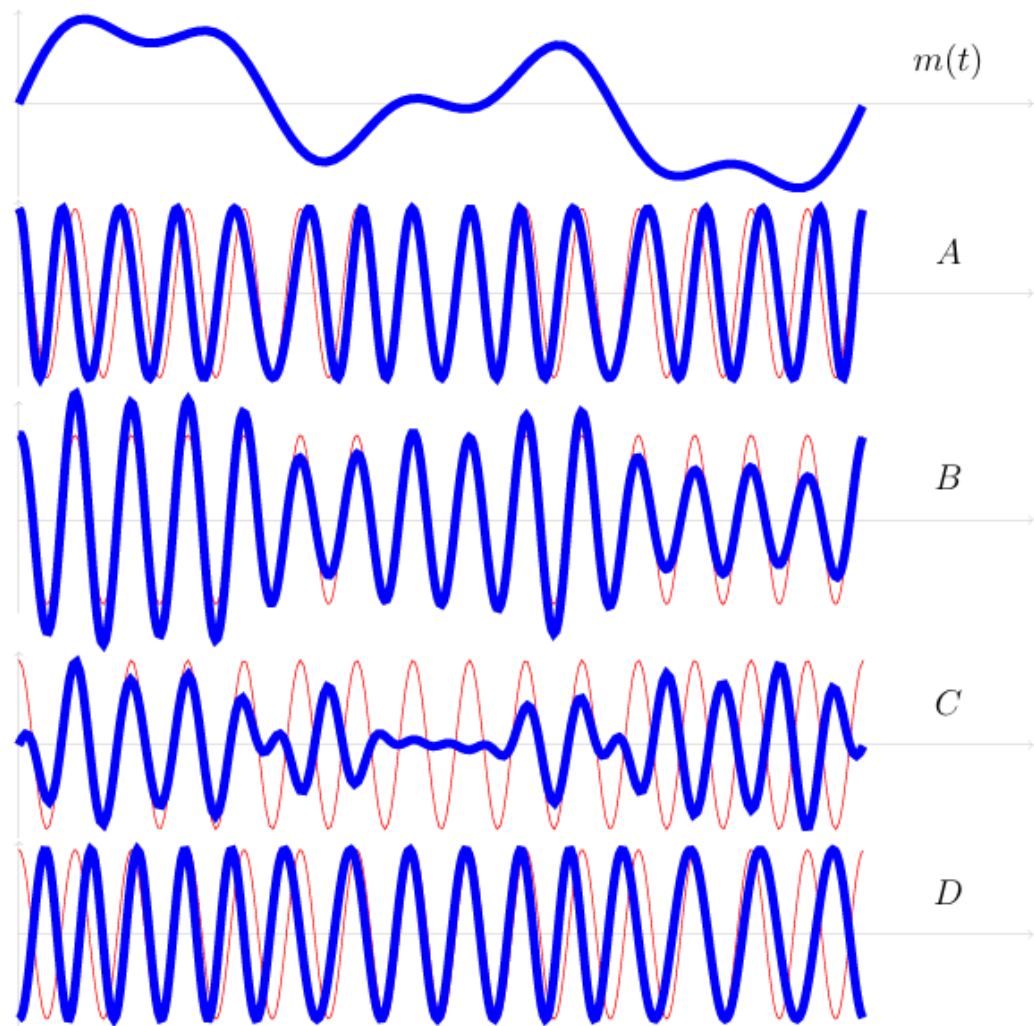


Figure 3: Four modulated signals.

Submission

Please submit your answers in a .pdf file named CS_HW4_studentID.pdf, replacing studentID with your actual student ID.